

General Physics II homework 1

Question 1. Consider the scalar field $f(x, y, z) = x^2ye^z + yz^3$ and the vector field $\vec{A}(x, y, z) = (xy + z^2)\hat{x} + (x^2 - yz)\hat{y} + (2xz + y^2)\hat{z}$.

Calculate the following quantities:

- 1) the gradient ∇f ;
- 2) the Laplacian $\nabla^2 f$;
- 3) the divergence $\nabla \cdot \vec{A}$;
- 4) the curl $\nabla \times \vec{A}$.

Then explicitly verify that $\nabla \times (\nabla f) = 0$ and $\nabla \cdot (\nabla \times \vec{A}) = 0$.

Question 2. Let φ and ψ be smooth scalar fields, and let $\vec{A}$ and $\vec{B}$ be smooth vector fields.

Verify the following vector identities:

- 1) $\nabla^2(\varphi\psi) = \varphi\nabla^2\psi + \psi\nabla^2\varphi + 2\nabla\varphi \cdot \nabla\psi$.
- 2) $\nabla \cdot (\varphi\nabla\psi - \psi\nabla\varphi) = \varphi\nabla^2\psi - \psi\nabla^2\varphi$.
- 3) $\nabla(\vec{A} \cdot \vec{B}) = (\vec{A} \cdot \nabla)\vec{B} + (\vec{B} \cdot \nabla)\vec{A} + \vec{A} \times (\nabla \times \vec{B}) + \vec{B} \times (\nabla \times \vec{A})$.

For part (3), verify the identity explicitly using Cartesian components.

Question 3. Consider the vector field $\vec{F} = x^2\hat{x} + y^2\hat{y} + z^2\hat{z}$.

Let V be the tetrahedral region in the first octant bounded by $x \geq 0$, $y \geq 0$, $z \geq 0$, $x + y + z \leq a$, where $a > 0$.

Check the divergence theorem

$$\iiint_V (\nabla \cdot \vec{F}) dV = \iint_{\partial V} \vec{F} \cdot d\vec{S}$$

by evaluating the two sides independently.

- 1) Calculate $\iiint_V (\nabla \cdot \vec{F}) dV$.
- 2) Calculate the flux of $\vec{F}$ through each of the four faces of the tetrahedron.
- 3) Show explicitly that the total outward flux agrees with the result in part (1).

Hint: On the inclined surface $x + y + z = a$, write $z = a - x - y$. For the outward orientation, $d\vec{S} = (\hat{x} + \hat{y} + \hat{z}) dx dy$.

Question 4. Consider the vector field $\vec{A} = -y\hat{x} + x\hat{y} + (x^2 + y^2)\hat{z}$.

Let S be the paraboloidal surface $z = 1 - x^2 - y^2$, $x^2 + y^2 \leq 1$, oriented upward.

The boundary C of S is the circle $x^2 + y^2 = 1$, $z = 0$, oriented counterclockwise when viewed from the positive z -axis.

Check Stokes' theorem

$$\oint_C \vec{A} \cdot d\vec{l} = \iint_S (\nabla \times \vec{A}) \cdot d\vec{S}$$

by calculating the two sides independently.

- 1) Parameterize the boundary C and calculate $\oint_C \vec{A} \cdot d\vec{l}$.
- 2) Calculate $\nabla \times \vec{A}$.
- 3) Calculate $\iint_S (\nabla \times \vec{A}) \cdot d\vec{S}$ and verify Stokes' theorem.

Hint: For a surface written as $z = z(x, y)$ with upward orientation,

$$d\vec{S} = \left(-\frac{\partial z}{\partial x}\hat{x} - \frac{\partial z}{\partial y}\hat{y} + \hat{z} \right) dx dy.$$

Question 5. The cylindrical coordinates (r, φ, z) are related to Cartesian coordinates by $x = r \cos \varphi$, $y = r \sin \varphi$, $z = z$.

Starting from the Cartesian coordinate system, derive the following results.

- 1) Show that $\frac{\partial}{\partial x} = \cos \varphi \frac{\partial}{\partial r} - \frac{\sin \varphi}{r} \frac{\partial}{\partial \varphi}$ and $\frac{\partial}{\partial y} = \sin \varphi \frac{\partial}{\partial r} + \frac{\cos \varphi}{r} \frac{\partial}{\partial \varphi}$.
- 2) Derive the gradient operator in cylindrical coordinates: $\nabla = \hat{e}_r \frac{\partial}{\partial r} + \hat{e}_\varphi \frac{1}{r} \frac{\partial}{\partial \varphi} + \hat{e}_z \frac{\partial}{\partial z}$.
- 3) Derive the Laplacian operator: $\nabla^2 = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \varphi^2} + \frac{\partial^2}{\partial z^2}$.
- 4) Using the result above, show that $f_n(r, \varphi) = r^n \cos(n\varphi)$ satisfies $\nabla^2 f_n = 0$ for $r > 0$ and any positive integer n .